

TOSONTSENGEL, Mongolia

WMO: 442250

Lat: 48.7631N

Lon: 98.2661E

Elev: 5653

StdP: 11.93

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-47.4	-43.1	-51.1	0.3	-46.5	-47.5	0.4	-43.1	17.7	1.1	13.6	-3.3	0.2	0	0.697

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.7	80.9	58.5	77.4	56.8	74.2	55.5	61.1	75.7	59.2	72.8	57.3	70.2	5.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.1	82.7	65.7	53.8	76.1	64.2	51.8	70.6	62.6	30.1	76.3	28.5	73.0	27.1	70.3	71.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.9	14.4	12.1	DB	-49.7	87.1	5.8	3.7	-53.9	89.8	-57.3	92.0	-60.6	94.1	-64.8	96.8	(4)
(5)			WB	-49.8	63.6	5.2	3.2	-53.6	65.9	-56.6	67.8	-59.5	69.5	-63.3	71.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	23.5	-23.8	-15.6	6.6	34.0	45.4	56.5	60.9	56.4	44.4	27.9	3.1	-16.6	(6)
(7)		DBStd	31.44	10.80	11.82	13.71	9.47	8.04	6.37	5.25	5.71	7.27	10.29	13.57	11.85	(7)
(8)		HDD50	10519	2289	1837	1345	482	184	18	1	13	194	685	1407	2064	(8)
(9)		HDD65	15194	2754	2257	1810	931	608	262	148	271	617	1150	1857	2529	(9)
(10)		CDD50	833	0	0	0	2	41	212	339	212	27	0	0	0	(10)
(11)		CDD65	31	0	0	0	0	0	6	21	4	0	0	0	0	(11)
(12)		CDH74	747	0	0	0	0	27	189	387	140	4	0	0	0	(12)
(13)		CDH80	139	0	0	0	0	1	26	92	20	0	0	0	0	(13)
(14)	Wind	WSAvg	3.2	1.2	1.7	3.4	5.4	5.5	4.0	3.3	3.0	3.6	3.3	2.1	1.6	(14)
(15)	Precipitation	PrecAvg	8.3	0.1	0.1	0.1	0.4	0.6	1.5	2.3	1.8	0.7	0.4	0.3	0.2	(15)
(16)		PrecMax	12.4	0.9	0.5	0.4	1.1	1.5	3.3	5.7	3.7	2.2	1.9	0.8	0.8	(16)
(17)		PrecMin	5.8	0.0	0.0	0.0	0.0	0.0	0.3	0.7	0.5	0.1	0.0	0.0	0.0	(17)
(18)		PrecStd	1.7	0.2	0.1	0.1	0.3	0.4	0.7	1.0	0.8	0.5	0.3	0.2	0.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.0	23.4	46.4	65.6	77.4	83.5	87.2	82.8	73.4	58.9	41.2	23.4	(19)
MCWB			12.0	20.0	36.0	46.7	53.3	58.6	63.2	59.2	53.5	43.3	33.7	20.4	(20)	
2%		DB	4.9	14.6	38.8	59.3	72.1	79.4	82.2	77.9	68.8	53.5	33.6	14.2	(21)	
		MCWB	3.5	12.4	31.5	43.1	51.7	57.0	60.0	57.1	51.6	40.6	27.9	12.5	(22)	
5%		DB	-2.2	8.4	33.6	54.8	67.7	75.6	78.1	74.1	64.4	49.0	28.2	7.7	(23)	
		MCWB	-3.1	7.0	28.3	40.9	48.9	55.4	58.1	55.9	48.9	38.0	24.1	6.5	(24)	
10%		DB	-8.0	2.8	28.5	50.6	63.2	71.9	74.6	70.5	60.2	44.5	22.8	1.4	(25)	
		MCWB	-8.6	1.6	24.6	38.8	46.3	53.8	56.7	54.6	46.9	35.3	19.8	0.5	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.6	20.3	36.5	48.2	56.9	61.1	65.8	62.0	55.2	44.4	34.7	20.6	(27)
MCDWB			13.7	23.3	45.8	62.9	74.8	78.3	83.0	77.0	70.0	57.2	41.8	23.6	(28)	
2%		WB	3.6	12.2	31.8	44.1	52.7	58.9	62.5	59.3	52.5	41.4	28.8	12.4	(29)	
		MCDWB	4.8	14.2	38.1	57.1	69.4	75.7	76.9	72.6	65.9	52.4	34.2	14.1	(30)	
5%		WB	-3.0	6.9	28.4	41.7	49.7	57.0	60.3	57.3	50.2	38.9	25.0	6.6	(31)	
		MCDWB	-2.2	8.4	33.4	53.6	65.8	71.8	73.6	69.9	62.4	49.0	29.3	7.8	(32)	
10%		WB	-8.5	1.6	24.5	39.1	47.1	55.2	58.4	55.7	47.7	36.2	21.1	0.6	(33)	
		MCDWB	-8.0	2.8	28.5	50.0	61.6	68.7	70.6	68.1	59.4	44.8	24.3	1.4	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	18.8	24.3	27.7	22.8	25.1	24.6	22.7	23.7	26.4	25.4	21.0	17.5	(35)
MCDBR			23.3	27.0	27.4	29.5	33.8	32.0	30.3	31.3	33.9	32.7	26.3	23.6	(36)	
MCWBR			22.2	24.8	21.0	17.5	18.7	15.8	13.7	14.9	19.3	20.9	20.4	22.1	(37)	
5% WB		MCDBR	23.5	26.9	26.9	28.1	31.4	27.5	25.8	26.2	30.9	31.7	25.3	23.5	(38)	
		MCWBR	22.3	24.8	21.0	17.4	18.7	14.7	13.3	13.2	18.1	20.9	20.3	22.0	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.221	0.245	0.286	0.325	0.330	0.334	0.337	0.319	0.286	0.267	0.245	0.226	(40)	
taud		2.343	2.295	2.265	2.321	2.345	2.432	2.459	2.479	2.521	2.494	2.407	2.335	(41)		
Ebn at Noon		279	293	295	293	295	293	291	292	294	284	266	260	(42)		
Edh at Noon		21	28	35	37	38	35	34	31	27	23	20	18	(43)		
(44)	All-Sky Solar Radiation	RadAvg	553	895	1389	1805	2043	2044	1933	1700	1426	950	565	433	(44)	
(45)		RadStd	28	40	56	73	122	95	107	70	73	54	23	18	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page